

Sartaaj Khan

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Education

University of Toronto

Sept. 2023 – Present

PhD Chemical Engineering and Applied Chemistry, Affiliated with Vector Institute

Principal Investigator: Dr. Seyed Mohamad Moosavi;

cGPA: 4.00/4.00

University of Toronto

Sept. 2018 – May 2023

BASc Chemical Engineering and Applied Chemistry, Minor in Bioengineering;

cGPA: 3.73/4.00

Research Experience

Doctoral Researcher

Sept. 2023 – Present

University of Toronto and Vector Institute for AI

Toronto, ON

Thesis: Learning the geometry of metal-organic frameworks (MOFs) using data-driven approaches

- Designed and implemented a multimodal machine learning framework that integrates powder X-ray diffraction (PXRD) patterns and textual descriptions of chemical precursors to link MOF synthesis pathways with targeted applications.
- Improved predictive performance through self-supervised learning and representation learning techniques, with fine-tuning on material properties including band gap, gas uptake (at multiple pressures), pore volume, and Henry's coefficient.
- Contributed to developing a novel graph-based representation (Labelled Quotient Graphs) to capture the bond topology of nanoporous materials, and benchmarked performance against leading graph neural network (GNN) architectures.
- Aided in the construction of a physics-informed machine learning model to predict dynamic viscosity of thermal fluids from small datasets and conducted uncertainty assessments of predictions.
- Leveraged insights from a large language model (LLM) to construct MOF-ChemUnity, a curated database linking experimental and computational MOF properties, to build ML models to predict water stability and discover MOFs for new applications.
- Mentored undergraduate and graduate students in topics related to machine learning in material science, informatics and representation learning.

Undergraduate Researcher

July 2021 – Sept. 2022

Advanced Membranes Laboratory, University of Toronto

Toronto, ON

- Developed a model showing the concentration profiles of common ions found in brackish water in a reverse osmosis process.
- Created a model involving different process modules in series such as a reverse osmosis membrane and a custom membrane specified.
- Used elemental-based modelling to compute the concentration profiles, flux of water and the flux of solute with respect to position and recovery ratio while including more complexities.
- Created and published a Python solution chemistry package called pySolution for the purpose of calculating parameters that are involved in non-ideal solutions from the Pitzer model.

Undergraduate Researcher

June 2021 – Sept. 2021

Ping Lab, University of California, Los Angeles

Los Angeles, CA

- Developed a bioinformatics tool for generating a more reliable protein sequence database for *sus scrofa*

cardiac proteomic studies.

- Researched and developed features relevant in proteomics such as sequence length, PE score and number of confirmed isoforms.
- Created a classifier trained against *homo sapien* protein data from UniProt and tested against *mus musculus* and *sus scrofa* protein data to determine “reliable” and “unreliable” proteins in those respective organisms.
- Constructed cardiovascular knowledge graphs and created link prediction algorithms to find unknown connections between proteins in different organisms – specifically in cardiac conduction and muscle contraction.

Undergraduate Researcher

Laboratory of Metabolic Systems Engineering, University of Toronto

May 2020 – Sept. 2020

Toronto, ON

- Created algorithms that genetically modify *E. coli* models and performed constrained optimization to optimize the flux of around 70 different industrially important chemicals through flux balance analysis.
- Modified mcPECASO simulator to obtain results with two-stage bioprocesses to analyze the impact of phenotypic switches on produced flux in iJO1366 and iML1515 *E. coli* models.

Publications

Updated as of 2025/07/20. For full list, please refer to my Google Scholar.

- **Khan, S. T.**, Moosavi, S. M. (2025). Connecting metal-organic framework synthesis to applications using multimodal machine learning. *Nature Communications*, 16(1), 5642.
- Ai, Q.*, **Khan, S. T.***, Barthel, S., Moosavi, S. M. (2025). “Capturing Global Features of Crystals from Their Bond Networks.” *AI for Accelerated Materials Design-ICLR 2025*.
- Song, Z., ..., **Khan, S. T.**, ..., & Duan, C. (2025). “Evaluating large language models in scientific discovery.” *arXiv preprint arXiv:2512.15567*.
- Jiang, H., **Khan, S. T.**, Werber, J. R., & Karanikola, V. (2025). Modeling scaling prevention and attainable recovery using hypothetical calcium-permeable reverse osmosis membranes. *ACS ES&T Engineering*, 6(1), 303–312.
- Kochi, M. R., Rezaei, H., **Khan, S. T.**, Mamillapalli, B., Ebrahimiazar, M., Ye, H., ... & Moosavi, S. M. (2025). Physics-informed machine learning for predicting temperature-dependent chemical properties. *Digital Discovery*.
- Pruyn, T. M., Aswad, A., **Khan, S. T.**, Huang, J., Black, R., & Moosavi, S. M. (2025). MOF-ChemUnity: Literature-Informed Large Language Models for Metal–Organic Framework Research. *Journal of the American Chemical Society*, 147(47), 43474–43486.
- Zimmermann, Y., Bazgir, A., ..., **Khan, S. T.**, ..., & Blaiszik, B. (2025). 32 examples of LLM applications in materials science and chemistry: Towards automation, assistants, agents, and accelerated scientific discovery. *Machine Learning: Science and Technology*, 6(3), 030701.
- Zimmermann, Y., ..., **Khan, S. T.**, ..., & Blaiszik, B. (2024). “Reflections from the 2024 large language model (LLM) hackathon for applications in materials science and chemistry.” *arXiv preprint arXiv:2411.15221*.

Invited Presentations

- **Khan, S. T.**, Moosavi, S. M. “Accelerated Discovery of Metal–Organic Frameworks through Synthesis-Informed Structure Generation.” [Poster]. Accelerate Conference 2025, Toronto, Canada
- Ai, Q., **Khan, S. T.**, Barthel, S., Moosavi, S. M. “Capturing Global Features of Crystals from Their Bond Networks.” [Paper]. AI for Accelerated Materials Design – ICLR, 2025
- **Khan, S. T.** “Unveiling insights of metal-organic frameworks using a self-supervised multimodal model with only PXRDs and precursors.” [Poster]. American Chemical Society (ACS), 2025, San Diego, California
- **Khan, S. T.**, Moosavi, S. M. “XRayPro: A Self-Supervised Multimodal Model for MOF Application Recommendations Using PXRD and Precursors.” [Paper]. AI for Accelerated Materials Design – NeurIPS, 2024, Vancouver, Canada.

- **Khan, S. T.** (2024, October 8). Unveiling Property Insights and Recommending MOF Applications from X-Ray Diffraction Patterns using Self-Supervised Transformer Models. [Presentation]. Canadian Society for Chemical Engineering (CSCHE) 2024, Toronto, Canada
- **Khan, S. T.**, Moosavi, S. M. (2024, August 7). “Using a self-supervised model for property predictions and MOF application recommendations with only powder x-ray diffraction patterns.” [Poster]. Accelerate Conference 2024, Vancouver, Canada

Work Experience

Artificial Intelligence Engineer

MGRID Energy Inc.

Sept. 2021 – Sept. 2022

Whitehorse, YK

- Created and tested a variety of object detection models (primarily YOLO) using convolutional neural networks for the purpose of vehicle classification.
- Designed the architecture of a hybrid nonlinear regression-convolutional neural network model for the automatic detection of wildfires.
- Researched and extracted different features to include in the wildfire models through literature done on previous machine learning models in the area.
- Generated a synthetic dataset for the purpose of training a natural language processing (NLP) model to send alerts on new projects for bidding.

Teaching Experience

Machine learning (ML) tutor

University of Toronto

Feb. 2025 – Present

Toronto, ON

- Served as a Machine Learning Tutor for the Data Science Institute (DSI)’s CrossTALK – a bootcamp focused on applying AI and laboratory validation techniques for drug discovery.
- Provided technical support by troubleshooting a range of issues, including syntax errors, memory constraints, data loading, and documentation, across machine learning frameworks such as XGBoost and LightGBM.
- Delivered conceptual instruction on machine learning models, clustering algorithms, and the selection of molecular fingerprints, helping students understand both theoretical foundations and practical applications.

Teaching Assistant (TA)

University of Toronto

Sept. 2025 – Present

Toronto, ON

Courses taught:

- CHE260: Thermodynamics and Heat Transfer
- CHE333: Chemical Reaction Engineering

Research Mentorship

- Zakariyya Brewster (Summer 2025): University of Toronto undergraduate student in Engineering Science; Common pitfalls in applying machine learning in metal-organic frameworks.
- Hari Om Chadha (Summer 2024): University of Toronto undergraduate student in Engineering Science; Pretraining graph convolutional networks using Barlow-Twin for predicting fluid properties.

Awards

- **Microscopy Society of America’s Award** (Feb 2026). Awarded by the Microscopy Society of America (\$500) for the submission of “VLRIMM - Vision-Language Retrieval for Identical Materials Morphology” for the Microscopy Hackathon 2026 for ranking as the best all-student team globally.
- **Canada Regional Award** (Feb 2026). Awarded by Elsevier Computational Material Science journal (\$500)

for the submission of “VLRIMM - Vision-Language Retrieval for Identical Materials Morphology” for the Microscopy Hackathon 2026 for ranking 1st in Canada.

- **Queen Elizabeth II/DuPont Canada Scholarship in Science and Technology** (Sept 2025 – Sept 2026). The Queen Elizabeth II Graduate Scholarship in Science and Technology (QEII-GSST) program is designed to encourage excellence in graduate studies in science and technology. Award of \$15,000 issued due to academic and professional achievements in research.
- **University of Toronto Fellowship** (Sept 2023 – Sept 2025). Received upon admission into the Master of Applied Science (MAsc) program at the University of Toronto in Chemical Engineering and Applied Chemistry for academic and professional achievements. Award of \$13,200 issued.
- **LLMs in Chemistry and Materials Hackathon** (May 2024). Awarded 3rd place (\$250 prize funded by Radical AI and Anthropic) in the LLM Hackathon for Applications in Materials and Chemistry for the development of PoreVoyant – a chemistry-informed AI agent that can generate new linkers to decrease the band gap in metal-organic frameworks while leveraging MOF literature. This work was featured in a Medium article.
- **Bayesian Optimization for Chemistry and Materials Hackathon** (March 2024). Awarded 2nd place (\$500 CAD prize funded by Acceleration Consortium) in the Bayesian Optimization for Chemistry and Materials Hackathon hosted by Acceleration Consortium at the University of Toronto. The project involved the application of Bayesian Optimization to accelerate the discovery of fluids with the highest heat transfer coefficients.
- **Dean’s List** (Sept. 2018 – May 2023). Award issued for achieving an average of at least an A- per semester with a full course load. This was achieved for all semesters of undergraduate.
- **NSERC USRA** (May 2020 – Sept 2020). Award of \$6000 CAD issued for completion of 16 weeks of research work for the summer 2020 term and high academic achievements.

Skills

Programming:	Python, MATLAB and Simulink, Cypher Query Language, Neo4j, HTML, CSS, Bash
Machine learning:	PyTorch, Tensorflow, Scikit-Learn
Software:	ChemDraw, AutoCAD, Aspen Plus, Aspen Dynamics, VESTA, iRASP
Applications:	Microsoft Word, Microsoft PowerPoint, Microsoft Excel, Microsoft VISIO